

DEV

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SUMMARY

I am Dev, a B.Tech Computer Science engineer specializing in AI/ML, working across cloud native infrastructure, distributed systems, blockchain protocol engineering, security, and full stack web development.

Selected as a **Google Summer of Code 2026 Contributor** at **Sugar Labs** for the **Linux Desktop OS Migration Project (GTK3 to GTK4)**. Accepted Letter

Core Contributor at **LFDT Web3j** (Linux Foundation Decentralized Trust), also selected as an **LFX** Linux Foundation Mentee for Ethereum protocol development and blockchain infrastructure engineering.

EXPERIENCE

Sugar Labs : Linux Desktop Environment

Google Summer of Code (GSoC) 2026 Contributor, Linux Desktop OS Migration

- Migrating core desktop environment components of a Linux based operating system from GTK3 to GTK4: Wayland compatibility layers, D-Bus lifecycle recovery, runtime GDK safety guards, and clipboard handling for system level stability, plus cross version compatibility fixes across GTK3/GTK4 and Wayland environments. #1106 (Core GTK4 Migration & Wayland Integration)
- **Merged PRs:** #1030, #1059, #1060, #1061, #1063 **Open PRs:** sugar-toolkit-gtk4 #33 #120

LFDT Web3j : Ethereum Protocol Engineering

Core Contributor, Member & LFX Linux Foundation Mentee

- Building Ethereum ABI encoding, transaction signing, the EIP-7594 blob transaction wrapper (Fusaka hardfork), websocket fallback systems, transaction pipeline correctness, JVM resource handling, and filter lifecycle management inside the core web3j codebase
- Fixed EIP-7594 cell proof encoding and blob wrapper serialization, EIP-4844 blob transaction signing, Utf8String ABI encoding, nested dynamic array decoding, invalid address recovery, and ABI struct code generation
- **Merged PRs:** #2293, #2289, #2284, #2263 (EIP-7594 Blob Wrapper), #2255, #2254 **PRs:** #2281, #2278, #2276, #2274, #2273, #2266

CNCF Koordinator : Kubernetes Scheduling Infrastructure

- Contributing to koordinator-sh, a CNCF sandbox project for QoS aware Kubernetes scheduling: coscheduling abstractions, scheduler queueing hints, CI reliability, and scheduler RBAC / webhook mutation logic
- **Open PRs:** #2997 (RFC: Multi-Scheduler Dispatcher & Conflict Arbitrator), #2969 (GangInfo abstraction for coscheduling), #2970 (queueing hints for core plugins), #2996 (NRI plugin annotations for CPUSet/GPUShare)

OpenSSF / Hyperledger Ecosystem

Security & Infrastructure Contributor

- Contributed to OpenSSF RSTUF (Repository Service for TUF) supply chain security infrastructure, Hyperledger Cacti, and Hyperledger Identus: container security, deployment hardening, blockchain interoperability tooling, SDK reliability, and CI workflows
- **Merged PRs:** RSTUF #968, Cacti #4183, #4180, Identus #591 **Open PRs:** RSTUF #960 (TUF metadata visualizer integration), Cacti #4176 (ERC6909 token support for SATP)

PROJECTS

BTC HashFrame: Bitcoin Encoding System

Live Demo

Next.js, TypeScript, Cryptography, Bitcoin Infrastructure

- Built a client side Bitcoin encoding platform for air gapped workflows supporting BIP39, BIP21, PSBT payloads, descriptors, and raw transaction QR generation. Designed a zero knowledge browser architecture with deterministic validation and secure export flows, ensuring sensitive wallet data never leaves the local device

NWC Tester: Lightning Integration Platform

Live Demo

Next.js, TypeScript, Lightning Network, JSON-RPC

- Developed a real time Nostr Wallet Connect (NWC) testing platform with Lightning Network integration, wallet command execution, transaction inspection, and monitoring workflows
- Integrated Bitcoin Connect and Alby SDK for secure wallet connectivity, invoice handling, payment execution, balance lookup, and responsive debugging dashboards

SHINRA LABS: AI Data Platform

Live Demo

React, Supabase, PostgreSQL, Distributed Workflow Systems

- Built a production style AI data labeling and dataset marketplace platform with scalable annotation pipelines, QA review systems, workforce routing, and realtime workflow management
- Integrated Supabase backend infrastructure with authentication, storage, realtime synchronization, and structured company/freelancer dashboards inspired by enterprise AI tooling platforms

TECHNICAL SKILLS

Languages: C,Java, Kotlin, TypeScript/JavaScript, Python, SQL, Bash

Web & Backend:React, Next.js, Node.js, Express.js, GraphQL, PostgreSQL, Supabase, REST APIs

Systems & DevOps: Linux, Distributed Systems,GTK3/GTK4, Docker, Kubernetes, CI/CD, GitHub Actions, Container Security

Blockchain: Ethereum, Web3j, JSON-RPC, Smart Contract Interaction, EVM, Bitcoin (BIP39/BIP21/PSBT), Lightning Network

AI, ML & Data: PyTorch, Prompt Engineering, LLM Integration, LangChain, Pandas